DISORDERS OF BLOOD FLOW AND BLOOD PRESSURE REGULATION



AGENDA

1. Analyze the pathophysiology and clinical manifestations of common disorders affecting blood flow and blood pressure regulation.
2. Apply the Clinical Judgment Model (CJM) to assess, diagnose, and develop appropriate nursing interventions for patients with cardiovascular disorders.



INTRODUCTION

- **Why is Blood Flow and Pressure Regulation Important?**
- The cardiovascular system is responsible for delivering oxygen and nutrients to tissues while removing metabolic waste. Any disruption in blood flow or abnormal blood pressure can lead to serious complications such as heart disease, stroke, organ damage, and even death.
- **Blood flow** is maintained by the heart's pumping action and the integrity of blood vessels.

1: DISORDERS OF BLOOD FLOW

- **Atherosclerosis**
- **Definition:**
- A progressive condition characterized by the accumulation of fatty plaques in the arterial walls, leading to narrowing and hardening of the arteries.

Atherosclerosis

- **Pathophysiology:**

1. **Endothelial injury:** Caused by high blood pressure, smoking, diabetes, or hyperlipidemia.
2. **Lipid accumulation:** LDL cholesterol deposits in the arterial walls.
3. **Inflammation:** Macrophages engulf LDL, forming foam cells and fatty streaks.
4. **Plaque formation:** Smooth muscle cells proliferate and form a fibrous cap over the fatty core.

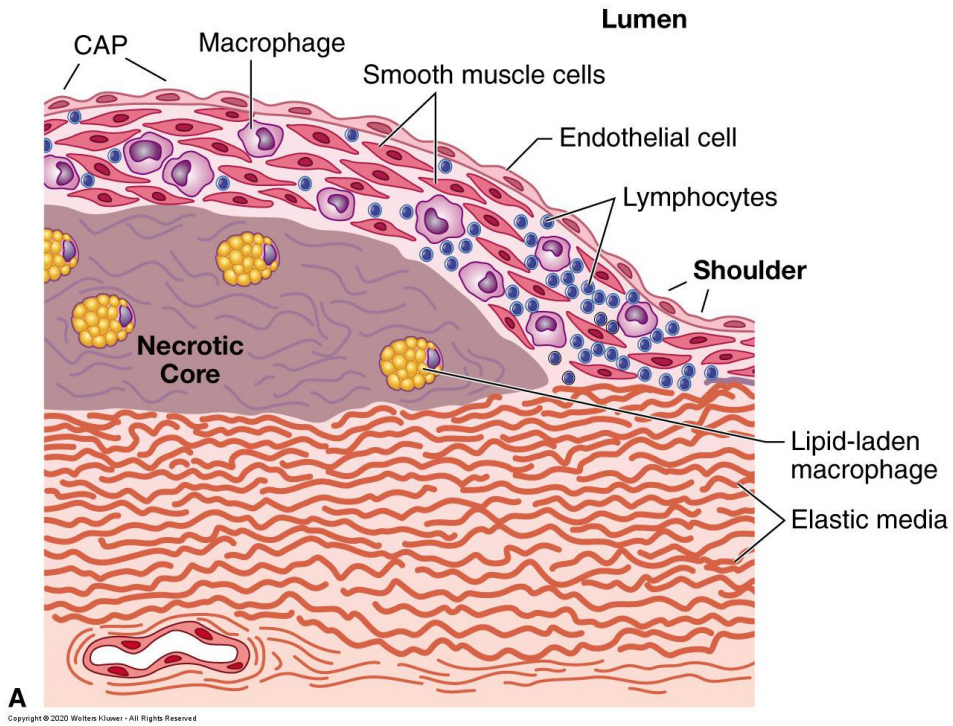
Atherosclerosis

1. **Complications:**
    1. Plaque rupture → thrombosis → myocardial infarction (heart attack) or stroke.
    2. Chronic narrowing → ischemia → angina or peripheral artery disease.

- **Risk Factors:**
- Modifiable: Smoking, diet, sedentary lifestyle, hypertension, diabetes.
- Non-modifiable: Age, genetics, male gender.

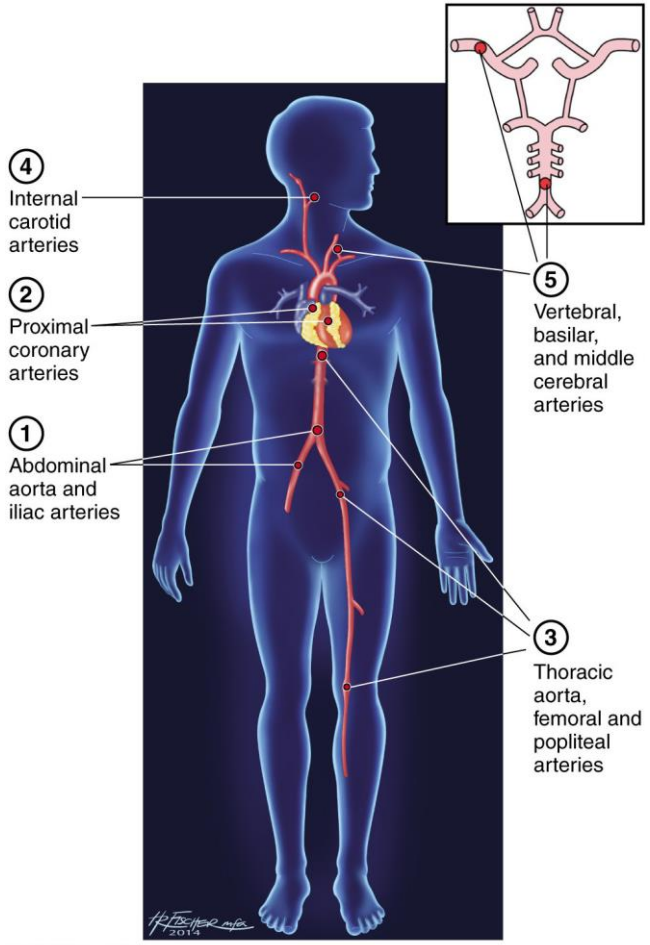
Atherosclerosis

- **Clinical Consequences:**
- **Coronary artery disease (CAD):** Chest pain, heart attack.
- **Cerebrovascular disease:** Stroke or transient ischemic attacks (TIA).
- **Peripheral artery disease (PAD):** Claudication, gangrene.
- https://youtu.be/i-uLFiyDyXk



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ANEURYSMS

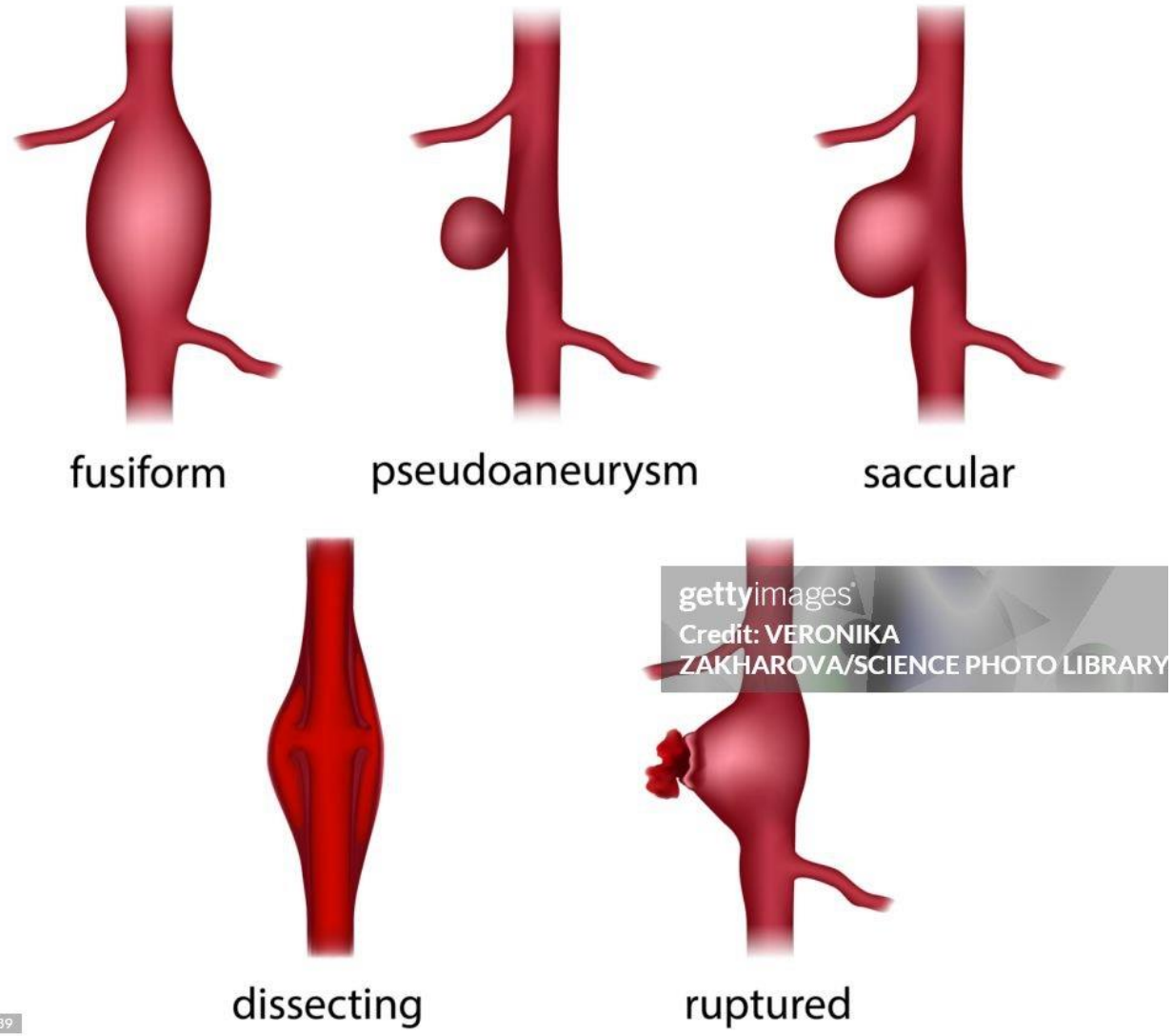
- **Definition:**
- A localized dilation or ballooning of an artery due to weakening of the vessel wall.
- **Types:**

1. **Saccular aneurysm:** Small, round outpouching.
2. **Fusiform aneurysm:** Entire arterial circumference is dilated.
3. **Dissecting aneurysm:** Blood enters the vessel wall through a tear, creating a false lumen.

ANEURYSMS

- **Causes:**
- Hypertension (weakens vessel walls).
- Atherosclerosis.
- Connective tissue disorders (e.g., Marfan syndrome).
- **Complications:**
- Rupture → internal bleeding → hypovolemic shock.
- Compression of nearby structures (e.g., brain aneurysm → neurological symptoms).

Types of aneurysm



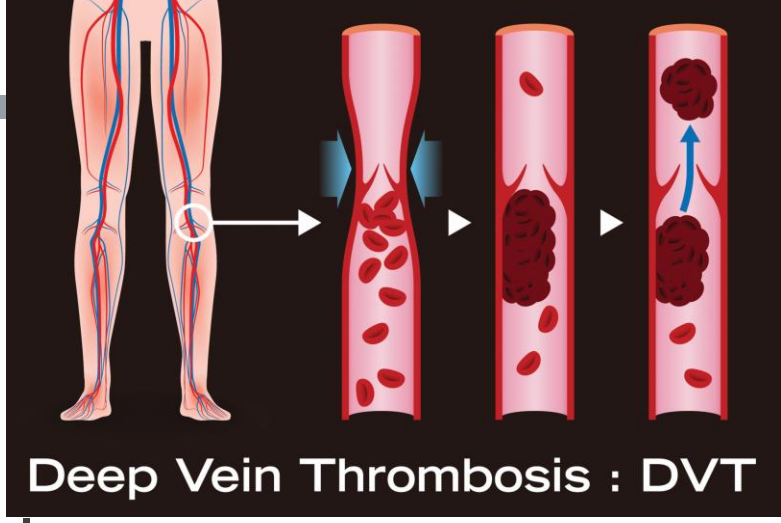
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VENOUS DISORDERS

- **Varicose Veins:**
- Enlarged, twisted veins due to incompetent valves leading to venous stasis.
- Common in pregnancy, prolonged standing, and obesity.
- Symptoms: Leg swelling, heaviness, aching pain.

DEEP VEIN THROMBOSIS (DVT):



- Formation of a blood clot in deep veins, often in the legs.
- **Virchow's Triad:**
  - **Venous stasis** (immobility, long flights).
  - **Hypercoagulability** (clotting disorders, pregnancy, cancer).
  - **Endothelial damage** (surgery, trauma).
- **Complication:** Pulmonary embolism (clot dislodges and travels to lungs, causing respiratory distress and death).

MICROCIRCULATION DISORDERS

- **Raynaud's Phenomenon:**
- Vasospasm of small arteries, causing pallor, cyanosis, and redness in fingers/toes.
- Triggered by cold or stress.
- Associated with autoimmune diseases (e.g., lupus, scleroderma).
- **Vasculitis:**
- Inflammation of blood vessels, leading to tissue ischemia.
- Causes: Autoimmune diseases, infections, drug reactions.
- Symptoms: Skin rashes, fever, joint pain.



III. DISORDERS OF BLOOD PRESSURE REGULATION

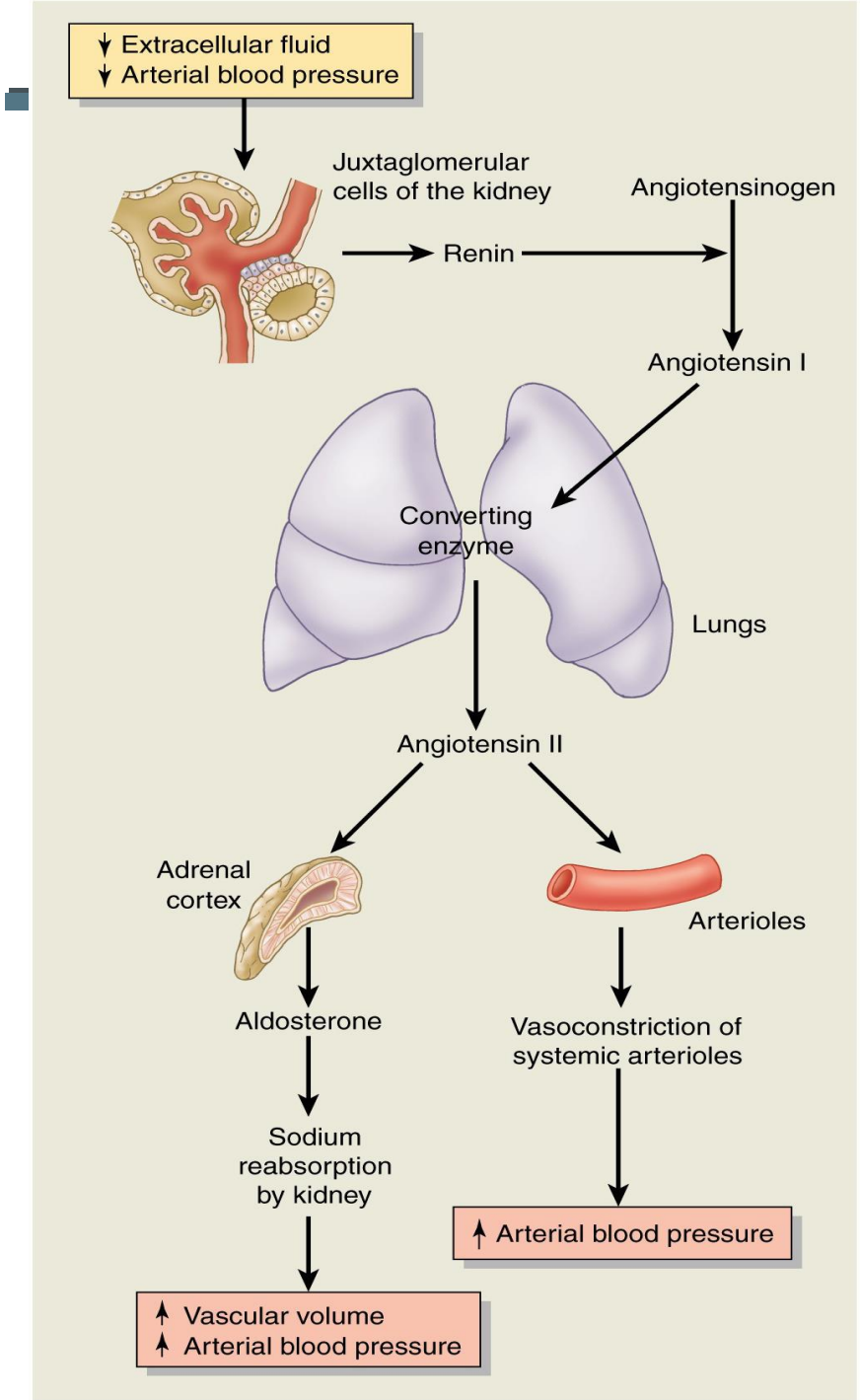
- Blood pressure depends on **cardiac output, systemic vascular resistance, and blood volume**.
- **1. Hypertension (HTN)**
- **Definition:**
- Persistent elevation of systolic BP ≥ 130 mmHg or diastolic BP ≥ 80 mmHg.

1. Hypertension (HTN), Types & pathophysiology

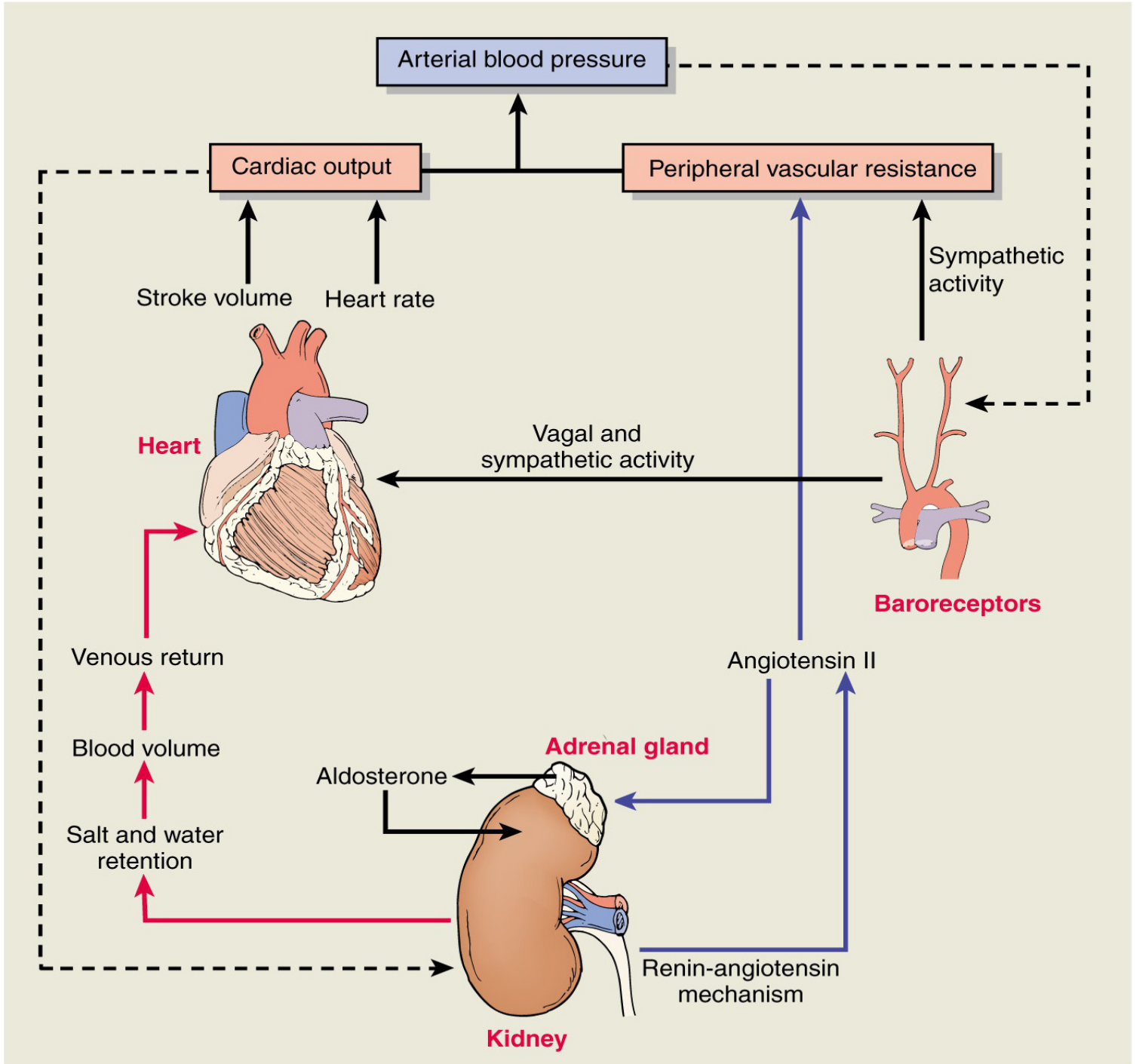
1. **Primary (Essential) Hypertension (90-95%)**
    1. No identifiable cause; linked to genetics, lifestyle, and aging.
2. **Secondary Hypertension (5-10%)**
    1. Due to an underlying condition (e.g., kidney disease, endocrine disorders, medication-induced).

- **Pathophysiology:**

- **Increased sympathetic nervous system activity:** Causes vasoconstriction and increased heart rate.
- **RAAS overactivation:** Leads to sodium retention and vasoconstriction.
- **Endothelial dysfunction:** Reduced nitric oxide (vasodilator), increased endothelin (vasoconstrictor).



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Hypertension (HTN)

- **Complications:**
- **Cardiac:** Left ventricular hypertrophy → heart failure.
- **Neurological:** Stroke, cognitive impairment.
- **Renal:** Chronic kidney disease.
- **Ophthalmic:** Hypertensive retinopathy → vision loss.

2. HYPOTENSION

- **Orthostatic Hypotension:**
- A drop in BP upon standing (systolic ↓ ≥ 20 mmHg or diastolic ↓ ≥ 10 mmHg).
- Causes: Dehydration, autonomic dysfunction (Parkinson's disease, diabetes).
- Symptoms: Dizziness, syncope.

2. HYPOTENSION

- A life-threatening condition of circulatory failure. **Types:**

1. **Hypovolemic Shock:** Due to blood/fluid loss (hemorrhage, dehydration).
2. **Cardiogenic Shock:** Pump failure (MI, heart failure).
3. **Distributive Shock:** Excessive vasodilation (sepsis, anaphylaxis, neurogenic).

- **Symptoms:** Hypotension, cold clammy skin, altered mental status.
- **Treatment:** IV fluids, vasopressors, oxygen

Diagnosis and Management

- **Diagnostic Tools:**
- **Blood pressure measurement:** Gold standard for hypertension diagnosis.
- **Lipid profile:** Identifies hypercholesterolemia.
- **ECG & Echocardiography:** Detects heart disease.
- **Angiography:** Evaluates arterial blockages.
- **Doppler ultrasound:** Assesses blood flow in veins and arteries.

Management Strategies

- **Lifestyle Modifications:**
- Diet: Low salt, DASH diet.
- Exercise: 30 min/day.
- Weight loss and smoking cessation.
- **Pharmacologic Therapy:**
- **Antihypertensives:** ACE inhibitors, beta-blockers, diuretics.
- **Statins:** Lower cholesterol, prevent atherosclerosis.
- **Anticoagulants:** Heparin, warfarin for DVT/PE.
- **Vasodilators:** Nitroglycerin for angina. https://youtu.be/JtBtk00EiVM

GROUP 1,2: **HYPERTENSIVE EMERGENCY**

- **Scenario:**
- Mrs. L, a **58-year-old female**, presents to the ED with a **severe headache, blurred vision, and nausea**. She has a history of **hypertension** but is **non-compliant with her medications**. Her **BP is 220/120 mmHg**, and she also reports **chest tightness and shortness of breath**.

- **Case 1: Hypertensive Emergency**

1. What are the **key clinical findings** in this patient that suggest a hypertensive emergency?
2. What **complications** could arise if this condition is not managed promptly?
3. What **immediate interventions** should be prioritized to lower BP safely?
4. How would you educate the patient about **long-term hypertension management** to prevent future emergencies?

GROUP 3,4: CARDIOGENIC SHOCK AFTER MYOCARDIAL INFARCTION

- **Scenario:**
- Mr. J, **67-year-old male**, with hypertension and **type 2 diabetes**, presents with **severe chest pain, sweating, and dizziness**. His BP is **85/50 mmHg**, HR **120 bpm**, RR **28**, SpO_2 **89%**. ECG shows **STEMI**.

1. What are the **clinical signs** that suggest the patient is in **cardiogenic shock**?
2. Why is **hypotension** concerning in this case, and how does it relate to cardiac output?
3. What are the **priority interventions** to stabilize this patient before PCI?
4. What are the **expected ECG and lab findings** in STEMI and cardiogenic shock?
5. How does diabetes affect **cardiovascular health** and increase the risk of complications?

GROUP 5,6: ACUTE DEEP VEIN THROMBOSIS (DVT) & PULMONARY EMBOLISM RISK

- **Scenario:**
- Mr. K, a **45-year-old male**, presents to the clinic complaining of **unilateral leg pain, swelling, and warmth** in his **left calf**. He recently **returned from a 10-hour international flight**. He denies trauma or previous DVT history but has a **family history of blood clots**. His vital signs are stable, but his **left calf is swollen, tender, and erythematous**.

- What are the risk factors for DVT in this patient?
- What assessment findings suggest DVT rather than another condition (e.g., cellulitis)?
- How would you explain the risk of pulmonary embolism to this patient?
- What nursing interventions and anticoagulation protocols should be initiated?
- How can the patient prevent future DVT episodes during trave

THANK YOU

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